BDM QUIZ-1

Complete Study Notes

Weeks 1 to 4 | IIT Madras BSc Data Science

Topics Covered

- Week 1: Economics Basics, Circular Flow, Agents, Government
- Week 2: Data Sources, Surveys, Sampling, Demographic Data
- Week 3: Demand, Supply, Elasticity, Equilibrium, Income Effects
- Week 4: Market Analysis, Porter's Five Forces, HHI, Industry Life Cycle
- Cheat Sheet: All Formulas & Quick Revision

WEEK 1: Economics Basics

1.1 What is an Economy?

An economy is a system where people produce, exchange and consume goods and services to satisfy their needs. Every country, city and household operates as some form of economy.

Microeconomics vs Macroeconomics

Aspect	Microeconomics	Macroeconomics
Focus	Individual units — one firm, one household	Whole economy
Studies	Pricing, firm behaviour, consumer choice	GDP, inflation, unemployment
Example	Why did Maggi raise its price?	Why is India's inflation at 6%?
Analogy	Zoom into one tree	Look at the entire forest

 Exam Tip: BDM is primarily microeconomics. Macro appears when discussing GDP, IIP, PMI.

The 5 Activities of an Economy

Activity	What it means	Indian Example
Production	Combining land, labour, capital to create goods/services	ITC producing biscuits
Exchange	Buying and selling through markets (Quid pro quo)	You buying biscuits at D-Mart
Distribution	How income is divided among contributors	Tata paying wages, dividends
Consumption	Households using goods/services to satisfy needs	You eating the biscuits
Investment	Using savings to build future productive capacity	Tata buying new machinery

Quid pro quo = 'This for that' — every exchange involves giving something to get something. Trade always creates value because both sides gain.

1.2 Agents in the Economy

Households

Households are consumers AND resource owners. They own the three factors of production:

- Land — natural resources
- Labour — their time and skills
- Capital — money, machines, equipment

Firms

Firms are producers. They combine land, labour and capital to produce goods/services and sell them. Firms include everything from street chai stalls to Tata Motors.

> **Exam Tip:** The same person can be both household (at home) and firm (running a business). Classify based on the role being described.

1.3 The Circular Flow Model

Basic Circular Flow

The economy operates as a loop between households and firms through two markets:

Market	What flows	Direction
Product Market	Goods & Services	Firms → Households
Product Market	Money (Expenditure)	Households → Firms
Factor Market	Land, Labour, Capital	Households → Firms
Factor Market	Wages, Rent, Profit	Firms → Households

Full Circular Flow — with Government & Foreign Sector

The complete model has 4 sectors:

- Households — consume, supply resources, pay taxes
- Firms — produce goods, hire resources, pay taxes
- Government — collects taxes, provides public goods, acts as producer AND consumer
- Foreign Sector — exports (injection) and imports (leakage)

Flow	Type
Government spending on roads	Injection
Taxes paid to government	Leakage
Exports to USA	Injection
Importing iPhones	Leakage

Household savings in bank	Leakage
Bank lending savings to firms	Injection

Leakage = money exiting the active spending loop (even temporarily). Injection = money re-entering the flow.

1.4 Intermediaries

Intermediaries sit between producers and consumers. All intermediaries are technically FIRMS.

Intermediary	Market	Indian Example
Wholesale stores	Product Market	Grain mandi in Punjab
Retail stores	Product Market	D-Mart, Kirana stores
Financial institutions	Factor Market	SBI — takes deposits, gives loans
Real estate companies	Factor Market	DLF leasing office space
Staffing companies	Factor Market	TeamLease placing workers at firms

1.5 Role of Government

3 Roles of Government

- Regulator — sets rules, collects taxes
- Producer — runs public sector enterprises (Indian Railways, ONGC, BSNL)
- Consumer — buys goods and services (defence ministry buying jets, offices buying computers)

4 Tools of Government Intervention

Tool	Effect on Flow	Indian Example
Taxation (GST, income tax)	Leakage	GST on every product purchase
Public spending	Injection	Building highways, IITs, AIIMS
Subsidies	Injection	LPG subsidy, PM Kisan, fertiliser subsidy
Regulation	Neither — controls behaviour	MSP for farmers, RBI repo rate, MRP

 Exam Tip: Fiscal policy = Govt taxes & spending. Monetary policy = RBI controlling interest rates and money supply.

1.6 Consumption Baskets

A consumption basket is the bundle of goods and services a typical household buys regularly. Categories include food, transport, housing, education, durables and leisure.

Engel's Law: As income rises, the proportion spent on food decreases, even though the absolute amount may increase. The freed share goes to durables, leisure, and aspirational categories.

WEEK 2: Data Sources & Surveys

2.1 Government Data Sources

Source	What it surveys	Key metric / note
Census of India	Every household — 100% coverage	Population, literacy, housing. Every 10 years.
Economic Census	Every firm/establishment	Employment, business type, rural vs urban
NSS / HCES	Sample of households	UMPCE — Usual Monthly Per Capita Expenditure
ASI (Annual Survey of Industries)	Registered factories	Industrial output, wages, raw material cost
PLFS (Periodic Labour Force Survey)	Sample of households	Employment and unemployment rates

UMPCE = Usual Monthly Per Capita Expenditure. It is the average amount one person in a household spends per month. Used to estimate market size.

2.2 Non-Government Data Sources

CMIE — Centre for Monitoring Indian Economy

Private research organisation. Runs the Consumer Pyramids Household Survey (CPHS) — most comprehensive private household survey in India.

- Tracks ~1,70,000 households
- Monthly updates — very frequent
- Captures: ownership data, consumption data, aspirational data
- Panel data — same households tracked over time
- PSU (Primary Sampling Unit) = villages and towns from 2011 Census
- USU (Ultimate Sampling Unit) = individual households from those villages
- Homogeneous Region = neighbouring districts with similar agro-climatic conditions, urbanisation, female literacy

Aspect	Government Sources	Non-Government (CMIE)
Coverage	Entire country — mandatory	Sampled — not compulsory
Frequency	Every 5-10 years	Monthly / quarterly

Cost	Free / publicly available	Paid / subscription
Bias	Low — no profit motive	May have commercial bias
Best for	Scale, breadth	Recency, depth, behaviour tracking

2.3 Types of Data

Type	What it is	Example
Cross-sectional	Many subjects at ONE point in time	Survey 1,000 households in January 2024
Time-series	ONE subject over MANY time periods	India GDP every year 2000-2024
Panel data	MANY subjects over MANY time periods	CMIE tracking same households every month for 5 years

Panel data = Cross-sectional + Time-series combined. It is the most powerful because you can track how the same subjects change over time.

2.4 Survey Concepts & Sampling

Key Terms

Term	Definition	Example
Population	The entire group you want to study	All households in Hyderabad
Sample	The smaller group you actually survey	1,000 households from Hyderabad
Sampling Unit	The individual thing you pick	One household = one sampling unit
Sampling Frame	The list from which you pick your sample	CMIE uses 2011 Census villages as frame

Types of Sampling

Method	Type	How it works	Bias risk
Simple Random	Probability	Pure lottery — every unit equal chance	Low
Stratified	Probability	Divide into groups, randomly pick from each	Low

Convenience	Non-probability	Survey whoever is easily accessible	High
Judgement	Non-probability	Expert decides who best represents population	Medium

Concept	Definition	Key difference
Sampling Error	Gap between sample result and true population value	Random, unavoidable, reduces with larger sample
Sampling Bias	Systematic misrepresentation due to how sample was picked	Systematic, avoidable, doesn't reduce with larger sample

Survey Biases

- Recall Bias — people can't accurately remember past behaviour. Fix: shorter recall periods or transaction records.
- Seasonal Bias — data collected at one time may not represent whole year. Fix: collect across multiple seasons or use annual averages.

2.5 Aspirational Data

Aspirational data tells businesses what people WANT to buy in the future, not what they currently buy.

Data Type	What it captures	Use
Ownership data	What do you currently own?	Understand current penetration
Consumption data	What did you spend money on this month?	Track spending patterns
Aspirational data	What do you intend to buy in next 1-2 years?	Estimate future market size

Aspiration Gap = difference between what people want and what they can afford. As income grows, the gap closes and aspirations become actual purchases — this is the future market opportunity.

WEEK 3: Demand, Supply & Market Concepts

3.1 Willingness to Pay & Demand

Willingness to Pay (WTP)

WTP = the maximum price a consumer is willing to pay for a good or service.

WTP is not fixed — it depends on how much you need or value the product at that moment (utility).

Formula: Consumer Surplus = WTP - Actual Price Paid

Law of Demand

As price increases, quantity demanded decreases. As price decreases, quantity demanded increases. (Ceteris paribus — everything else constant.)

Movement vs Shift of Demand Curve

Type	Cause	What happens to curve	Example
Movement along curve	Price of the SAME good changes	Stays in same position, move up/down	Maggi price drops — buy more Maggi
Shift of entire curve	Non-price factor changes	Whole curve moves left or right	IPL season — more demand for TVs at same price

Factors that SHIFT Demand Curve

Factor	Effect	Example
Income increases	Shifts right (normal good)	Salary hike — buy more restaurants
Income decreases	Shifts left	Job loss — spend less
Substitute gets cheaper	Shifts left	Pepsi price drops — less Coke bought
Complement gets cheaper	Shifts right	Petrol cheaper — more cars bought
Tastes change	Either direction	Health trend — more demand for oats

3.2 Normal Goods vs Inferior Goods

Type	Definition	Income up effect	Indian Examples
Normal Good	Demand increases when income increases	Demand shifts RIGHT	Restaurants, cars, smartphones, flights
Inferior Good	Demand DECREASES when income increases	Demand shifts LEFT	Bus travel, coarse grains, generic brands
Luxury Good	Demand grows faster than income (YED > 1)	Demand shifts far RIGHT	Foreign holidays, luxury cars, designer clothes
Giffen Good	Extreme inferior — price up = buy more	Special case	Very cheap staples for extremely poor households

 Exam Tip: Inferior does NOT mean low quality. It is a purely economic term meaning demand falls when income rises. Bus travel is not low quality — people just switch to cars when richer.

3.3 Elasticity of Demand

Price Elasticity of Demand (PED)

Measures how sensitive quantity demanded is to a change in its own price.

Formula: PED = % Change in Quantity Demanded / % Change in Price

PED Value	Type	Meaning	Example
Greater than 1	Elastic	Demand very sensitive to price	Pepsi — switch to Coke if price rises
Less than 1	Inelastic	Demand not sensitive to price	Petrol — need it regardless of price
Equal to 1	Unit elastic	Demand changes exactly as much as price	-
Equal to 0	Perfectly inelastic	Demand doesn't change at all	Insulin for diabetics

What determines PED?

- More substitutes available → more elastic
- Necessity → inelastic. Luxury → elastic
- Longer time horizon → more elastic (people find alternatives)
- Habit/addiction → less elastic (e.g. cigarettes)

Income Elasticity of Demand (YED)

Formula: YED = % Change in Quantity Demanded / % Change in Income

YED Value	Type of Good	Example
Positive > 1	Luxury good	Foreign holidays
Positive 0 to 1	Normal necessity	Rice, basic clothing
Negative	Inferior good	Bus travel, coarse grains

Cross-Price Elasticity of Demand (XED)

Formula: XED = % Change in Qty Demanded of Good A / % Change in Price of Good B

XED Value	Relationship	Example
Positive	Substitutes	Swiggy price up → more Zomato orders
Negative	Complements	Petrol price up → fewer cars bought
Zero	Unrelated goods	Tea and cement

3.4 Supply & Market Equilibrium

Law of Supply

As price increases, quantity supplied increases. As price decreases, quantity supplied decreases. (Ceteris paribus)

Factors that Shift Supply Curve

Factor	Effect on Supply	Example
Input costs increase	Shifts left	Wheat prices rise — bread makers supply less
Technology improves	Shifts right	New machine — produce more at same cost
More producers enter	Shifts right	More farmers grow tomatoes
Government subsidy	Shifts right	Fertiliser subsidy — farmers produce more
Natural disaster	Shifts left	Floods destroy crops

Market Equilibrium

Equilibrium = the price at which quantity demanded = quantity supplied. No shortage, no surplus. Market naturally gravitates here.

Situation	Name	What happens next
Price above equilibrium	Surplus (supply > demand)	Sellers drop prices to clear stock
Price below equilibrium	Shortage (demand > supply)	Buyers bid prices up
Price at equilibrium	Equilibrium	No pressure to change

 Exam Tip: When BOTH demand and supply increase: Quantity definitely increases. Price direction is AMBIGUOUS — depends on which shift is larger.

WEEK 4: Market Analysis & Industry

4.1 Estimating Market Size

3-Step Framework

- Step 1: Define your target population — use Census and demographic data
- Step 2: Apply filters — income level, geography, awareness, aspiration (CMIE data)
- Step 3: Multiply remaining population x average spend per person (UMPCE)

Formula: Market Size = Number of potential buyers x Average spend per buyer

Term	Full Form	Meaning
TAM	Total Addressable Market	Everyone who could ever buy — the dream
SAM	Serviceable Addressable Market	Those you can realistically reach
SOM	Serviceable Obtainable Market	Your realistic share of the market

 Exam Tip: TAM > SAM > SOM. Investors laugh at startups that claim TAM = every person in India. Be realistic with filters.

4.2 IIP & PMI

Aspect	IIP	PMI
Full form	Index of Industrial Production	Purchasing Managers Index
Published by	MOSPI (Government)	S&P Global (Private)
Frequency	Monthly	Monthly
Method	Actual output data from factories	Survey of purchasing managers
Scale	Base year 2011-12 = 100	Above 50 = expansion, Below 50 = contraction
Speed	Lags by ~6 weeks	Released within days of month-end
Nature	What actually happened (lagging)	Business sentiment (leading indicator)
Covers	Manufacturing 78%, Mining 14%, Electricity 8%	Manufacturing PMI + Services PMI

PMI is a LEADING indicator — predicts what will happen. IIP is a LAGGING indicator — confirms what happened. Use both together for full picture.

4.3 Porter's Five Forces

A framework to analyse how competitive and profitable an industry is. Higher the combined force = lower the profitability.

Force	Question it answers	High force means	Example
Industry Competition	How intensely do existing players compete?	Price wars, low margins	Airlines — IndiGo vs Air India vs SpiceJet
Threat of New Entrants	How easy to enter the industry?	Market share threatened	Jio entering telecom and wiping out rivals
Supplier Power	Can suppliers dictate terms?	Higher input costs	Boeing/Airbus supplying aircraft to all airlines
Buyer Power	Can buyers negotiate prices down?	Forced discounts	Walmart forcing FMCG companies to give deep discounts
Threat of Substitutes	Can buyers switch to different product?	Price ceiling on your product	Trains as substitute for flights

 Exam Tip: Substitutes are DIFFERENT products solving the same need. Competitors are same product, same industry. Google Maps ordering food = substitute for Zomato, not a competitor.

Industry	Attractiveness	Why
Pharmaceuticals	High (attractive)	High entry barriers, few substitutes, many suppliers
Airlines	Low (unattractive)	All 5 forces high — price wars, powerful suppliers (Boeing/Airbus), easy substitutes

4.4 Market Concentration — CR & HHI

Concentration Ratio (CR)

Measures the combined market share of the top N firms in an industry.

Formula: CRn = S1 + S2 + S3 + ... + Sn (sum of top N firms' market shares)

CR4 Value	Market Structure

0% - 40%	Competitive market
40% - 60%	Moderately concentrated
60% - 80%	Highly concentrated
80% - 100%	Near monopoly / oligopoly

HHI — Herfindahl-Hirschman Index

More powerful than CR — squares market shares to give more weight to dominant firms.

Formula: HHI = $S1^2 + S2^2 + S3^2 + ... + Sn^2$ (ALL firms in industry, not just top N)

HHI Value	Market Structure
Below 1,500	Competitive market
1,500 - 2,500	Moderately concentrated
Above 2,500	Highly concentrated
10,000	Pure monopoly (one firm = 100%)

Why HHI is better than CR: CR treats all top firms equally. HHI penalises dominance by squaring. A market with one firm at 90% looks same as four equal firms in CR4 = 100%, but HHI clearly shows the dominance. CCI uses HHI to evaluate mergers.

4.5 Industry Classification — NIC Codes

System	Full Name	Used by
NIC	National Industrial Classification	India (MOSPI)
ISIC	International Standard Industrial Classification	United Nations — global standard
NACE	European Classification	Europe

NIC Code Hierarchy (5 levels)

Level	Digits	Example
Section	1 letter	C = Manufacturing
Division	2 digits	10 = Food products
Group	3 digits	107 = Bakery products
Class	4 digits	1071 = Bread, biscuits, cakes

Sub-class	5 digits (most specific)	Most detailed breakdown

4.6 Industry Life Cycle

Stage	Growth	Competition	Profit	Strategy	Indian Example
Introduction	Slow	Very few players	Low/negative	Build awareness, invest	EV industry 2024
Growth	Rapid	Many entrants	Rising	Grab market share fast	OTT streaming 2018-2022
Maturity	Stable	Consolidation	High but stable	Efficiency, cost cutting	Telecom, Cement
Decline	Negative	Players exiting	Falling	Exit or pivot	Print newspapers

 Exam Tip: Best time to enter = end of Introduction / early Growth stage. Maturity = most profitable but hardest to gain share. Never invest in Decline without a pivot plan.

4.7 Pivot Tables & Data Visualisation

Pivot Table — 4 Components

Component	What it does	Example
Rows	Groups data vertically	State names down the left
Columns	Groups data horizontally	Months across the top
Values	The number being calculated	Sum of sales, Average price
Filters	Restricts which data is shown	Only show South India data

Chart Type Guide

Chart Type	Use when	Example
Pie / Donut	Showing parts of a whole (% share)	Market share of telecom players
Bar / Column	Comparing categories	Sales by state
Line chart	Showing trends over time	Monthly IIP growth 2020-2024
Scatter plot	Relationship between two variables	Income vs spending
Histogram	Distribution of data across ranges	Household income distribution

Stacked bar	Composition + comparison across categories	Category-wise sales by quarter
Pareto chart	Bar (descending) + Line (cumulative) — 80/20 rule	Top medication errors causing 80% of problems

Pareto Principle (80-20 rule): 80% of effects come from 20% of causes. Pareto chart shows this — bars in descending order, line showing cumulative %.

QUICK REVISION CHEAT SHEET

All Formulas, Key Definitions & Exam Tips

All Formulas

Formula	Expression	Key interpretation
Consumer Surplus	WTP - Actual Price	Higher CS = customer feels great value
Price Elasticity (PED)	% Change in Qty / % Change in Price	>1 elastic, <1 inelastic
Income Elasticity (YED)	% Change in Qty / % Change in Income	+ve normal, -ve inferior, >1 luxury
Cross-Price Elasticity (XED)	% Change in Qty of A / % Change in Price of B	+ve substitutes, -ve complements
Concentration Ratio (CR)	S1 + S2 + ... + Sn	>80% = near monopoly
HHI	$S1^2 + S2^2 + ... + Sn^2$	>2500 highly concentrated, max=10000
Market Size	No. of buyers x Avg spend per buyer	TAM > SAM > SOM
UMPCE	Total HH expenditure / (HH size x months)	Per person per month spending

Key Definitions — One Liners

Term	One-line definition
Quid pro quo	Something for something — every voluntary trade creates value for both sides
WTP	Maximum price a consumer is willing to pay
Ceteris paribus	Everything else remaining constant — condition for law of demand/supply
Equilibrium	Price where quantity demanded = quantity supplied — no surplus, no shortage
Surplus	Supply > demand — price too high, will fall
Shortage	Demand > supply — price too low, will rise
Normal good	Demand increases when income increases
Inferior good	Demand decreases when income increases (NOT low quality)
Giffen good	Extreme inferior — price up, demand also up (very rare)

Veblen good	Luxury — higher price makes it MORE desirable (status symbol)
Panel data	Same subjects tracked over multiple time periods
UMPCE	Usual Monthly Per Capita Expenditure — avg spend per person per month
Sampling error	Random gap between sample and true population — unavoidable
Sampling bias	Systematic misrepresentation — avoidable, doesn't reduce with larger sample
Recall bias	People can't accurately remember past behaviour
Seasonal bias	Data from one season doesn't represent whole year
PMI	Above 50 = expansion, below 50 = contraction of business activity
IIP	Measures actual industrial output vs base year (2011-12 = 100)
HHI	Sum of squared market shares. >2500 = highly concentrated. Max = 10,000
Aspiration gap	Difference between what people want and can afford — future market potential
Engel's Law	As income rises, % spent on food decreases
CR4	Combined market share of top 4 firms in an industry

Data Sources — Quick Reference

Source	Type	What it tells you	Frequency
Census	Govt	Total population, literacy, housing	Every 10 years
Economic Census	Govt	All firms — employment, type	Every few years
NSS / HCES	Govt	Household spending (UMPCE)	Every few years
ASI	Govt	Factory output, wages	Annual
PLFS	Govt	Employment & unemployment	Quarterly
CMIE-CPHS	Private	Consumption + aspirations (panel)	Monthly

Porter's Five Forces — Quick Guide

Force	High = Bad because...	Low = Good because...
Industry competition	Price wars eat margins	Can maintain profitable pricing
New entrants threat	Market share shrinks	Barriers protect existing players
Supplier power	Input costs forced up	Can negotiate better prices
Buyer power	Forced to give discounts	Can maintain pricing

Substitute threat	Caps your price ceiling	Customers have no alternative

Industry Life Cycle — Quick Guide

Stage	Key signal	Indian Example 2024
Introduction	Slow growth, heavy investment, few players	Electric vehicles
Growth	Rapid expansion, many new entrants	Quick commerce (Blinkit, Zepto)
Maturity	Stable, consolidation, efficiency focus	Telecom, FMCG
Decline	Shrinking market, players exiting	Print newspapers, landlines

Sampling Types — Quick Guide

Method	Type	Bias	When to use
Simple random	Probability	Low	Small, homogeneous population
Stratified	Probability	Low	Diverse population (like India!)
Convenience	Non-probability	High	Exploratory research only
Judgement	Non-probability	Medium	Expert knowledge of population

Chart Types — Quick Guide

If asked about...	Use this chart
Market share / parts of whole	Pie chart
Comparing categories / groups	Bar chart
Trend over time	Line chart
Relationship between two variables	Scatter plot
Distribution across ranges	Histogram
Composition + comparison	Stacked bar chart
80-20 rule / prioritisation	Pareto chart

Top 10 Exam Traps to Avoid

Trap	Correct understanding

Inferior good = low quality product	NO — inferior = demand falls when income rises. Bus travel is not low quality!
Trade creates value only for seller	NO — both sides gain in voluntary trade (quid pro quo)
Saving in FD is always good for economy	Saving = leakage. Only becomes injection when bank lends it to firms.
PMI > IIP for quick decisions	YES — PMI released within days, IIP lags 6 weeks
CR4 = 100% means monopoly	NO — could be 4 equal firms. HHI reveals actual concentration.
Swiggy and Zomato are substitutes	NO — they are direct COMPETITORS. Substitutes = different product, same need.
Both D and S increase = price increases	WRONG — price is ambiguous. Quantity definitely increases.
Google Maps ordering food = competitor of Zomato	NO — substitute (different product solving same need)
Seasonal survey data represents whole year	NO — seasonal bias. Need to survey across multiple seasons.
Sampling error can be eliminated	NO — only reduced with larger sample. Bias can be eliminated.

All the best for your Quiz! You've got this!

Remember: Understand the concept, don't just memorise. Apply to examples.